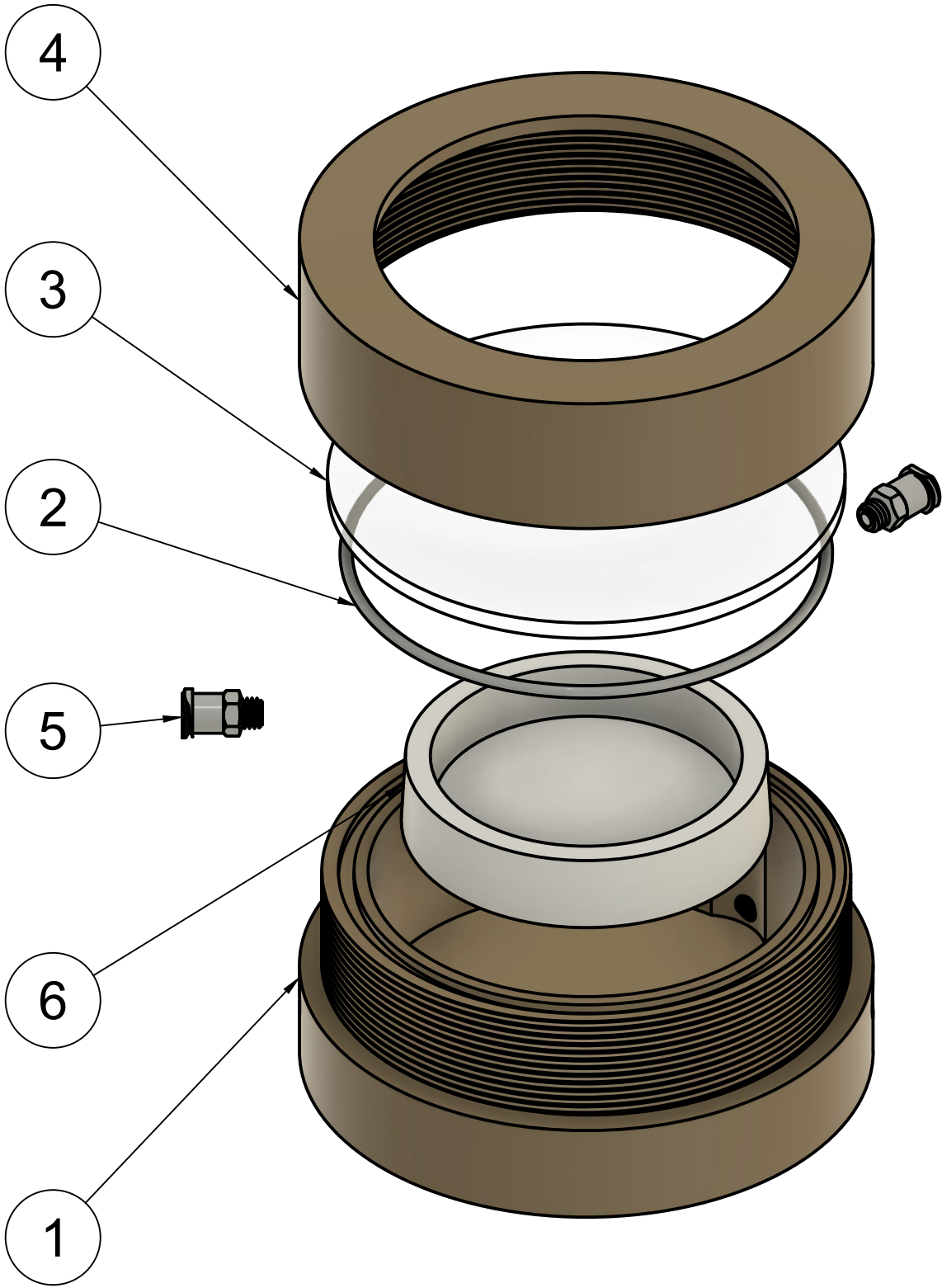


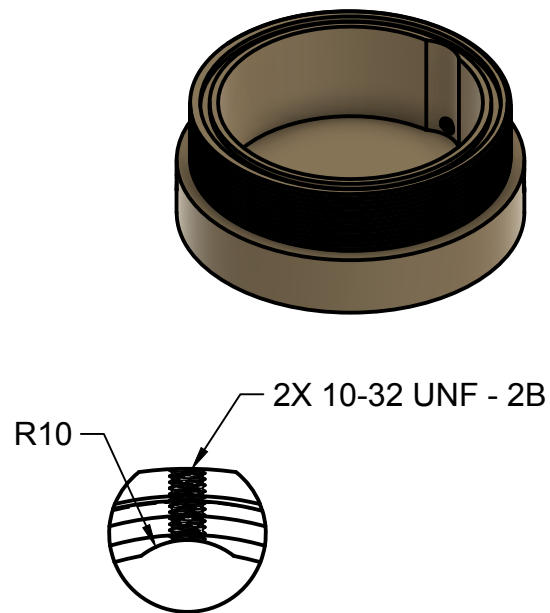
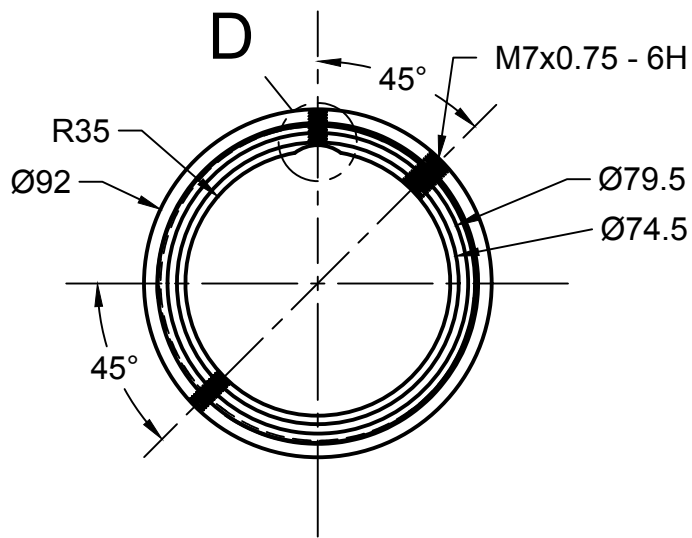
Cell Blast Assembly

2D Drawing

PARTS LIST				
ITEM	QTY	PART NUMBER	MATERIAL	MASS
1	1	CBA BOTTOM	PEEK	134.257 G
2	1	O-RING	RUBBER, BUTYL	0.091 G
3	1	CBA TOP COVER	POLYCARBONATE, CLEAR	19.478 G
4	1	CBA RIM	PEEK	42.511 G
5	2	LUER INSERT	POLYCARBONATE, CLEAR	0.01 OUNCEMASS
6	1	PDMS SUBSTRATE	ABS PLASTIC	17.009 G

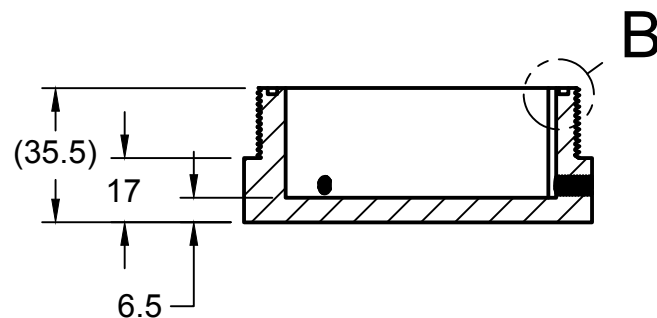
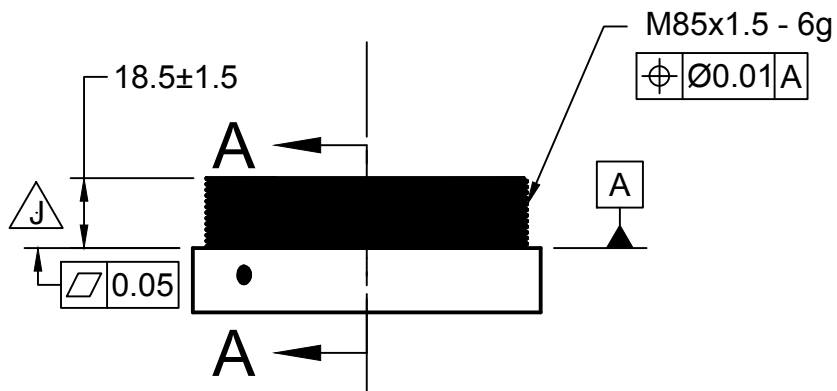


		PROJECT			
		BE-GY 6523 CAD Project			
		TITLE			
		Cell Blast Apparatus Bill of Materials			
APPROVED		SIZE	CODE	DWG NO	REV
CHECKED	Carolina Cen 4/21/25	B	ASME Y14.5	CBA-001	N/A
DRAWN	Saba Khan 4/18/25	SCALE 1:1		WEIGHT 241.696 G	SHEET 1/7

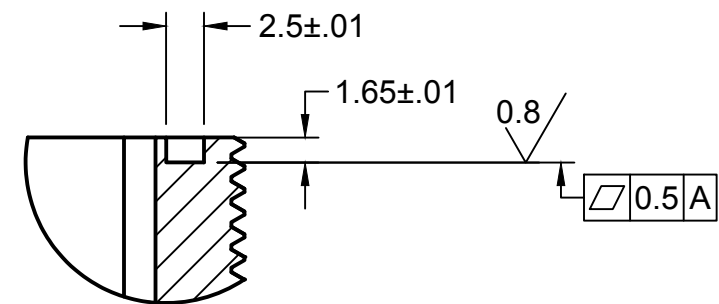


DETAIL D
SCALE 1:1

REVISION HISTORY			
REV	DESCRIPTION	DATE	APPROVED
A	1:0.9 SCALE DOWN	4/03/25	SK, CC
B	1:0.75 SCALE DOWN	4/17/25	SK, CC
C	THREADED ENGAGEMENT LENGTH INCREASED FROM 10 MM TO 13 MM	4/17/25	SK, CC
D	OUTLET HOLE POSITION MOVED 45 DEGREES	4/17/25	SK, CC
E	M7X0.75 - 6H THREADS ADDED TO PRESSURE SENSOR HOLE	4/17/25	SK, CC
F	PRESSURE SENSOR HOLE POSITION MOVED 45 DEGREES	4/17/25	SK, CC
G	MATERIAL CHANGED FROM HDPE TO PPSU	4/17/25	SK, CC
H	O-RING GROOVE WIDTH AND DEPTH ADJUSTED	4/17/25	SK, CC
J	THREAD ENGAGEMENT LENGTH INCREASED FROM 13 MM TO 18.5 MM	4/21/25	SK, CC



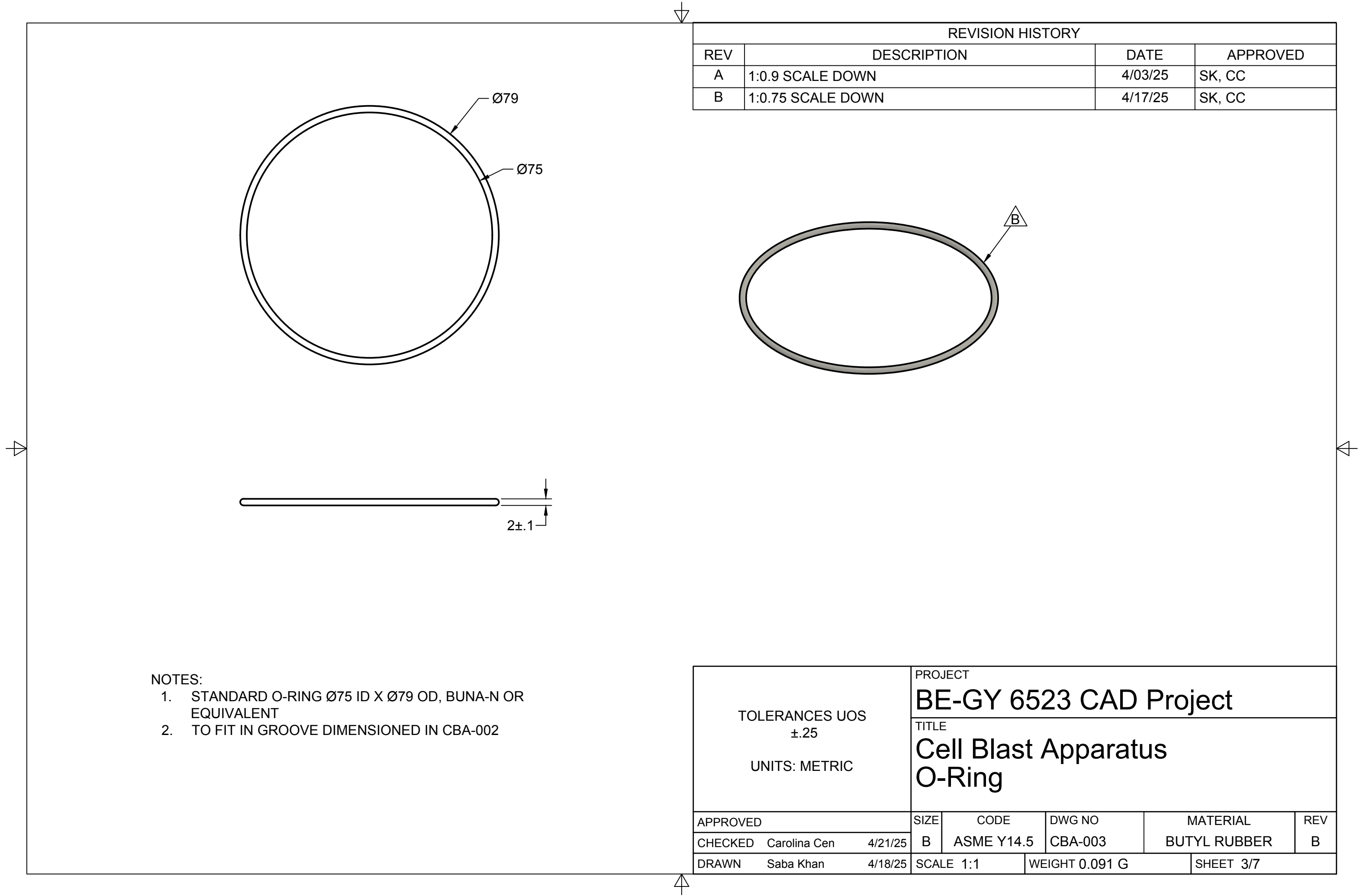
SECTION A-A
SCALE 1:2

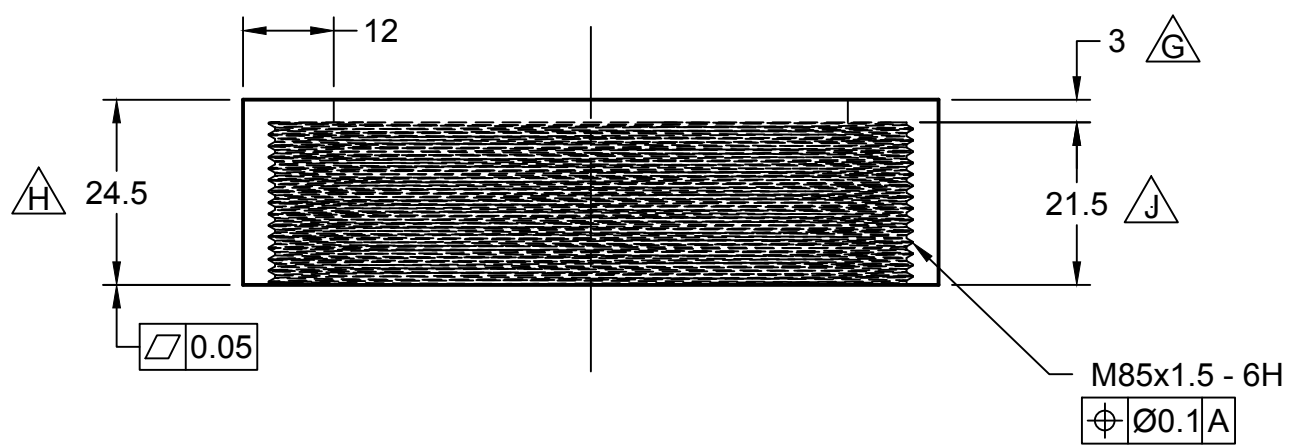
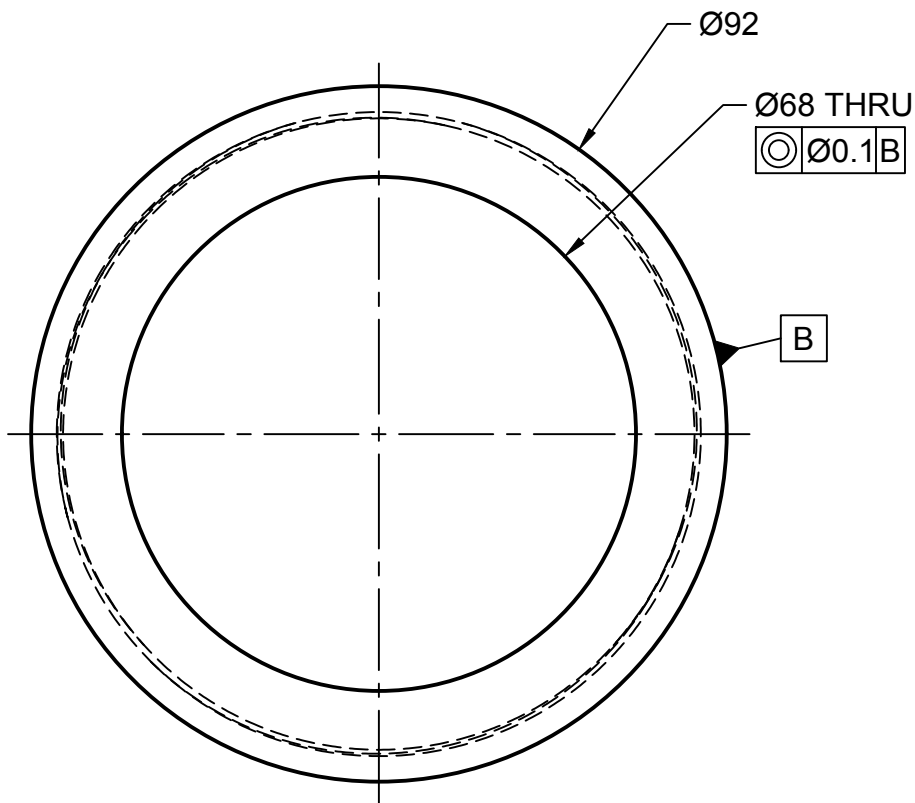


DETAIL B
SCALE 2:1

- NOTES:
- DETAIL B OUTLINES THE DIMENSIONS FOR THE STANDARD O-RING DETAILED IN CBA-003.
 - REMOVE ALL BURRS AND SHARP EDGES.
 - STANDARD 10-32 UNF THREAD SIZE PC LUER COUPLINGS TO BE INSERTED IN 10-32 UNF - 2B HOLES

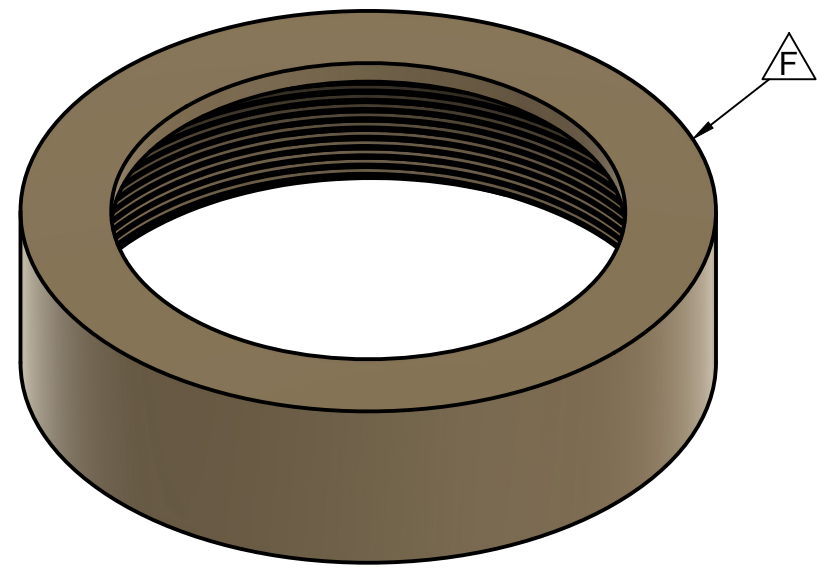
TOLERANCES UOS 1PL ±1 2PL ±.05 TOL ON ANGLE ±1° UNITS: METRIC			PROJECT BE-GY 6523 CAD Project			
			TITLE Cell Blast Apparatus CBA Bottom			
APPROVED	SIZE	CODE	DWG NO	MATERIAL	REV	
CHECKED Carolina Cen 4/21/25	B	ASME Y14.5	CBA-002	POLYPHENYLSULFONE (PPSU)	J	
DRAWN Saba Khan 4/19/25	SCALE 1:2		WEIGHT 134.257 G		SHEET 2/7	





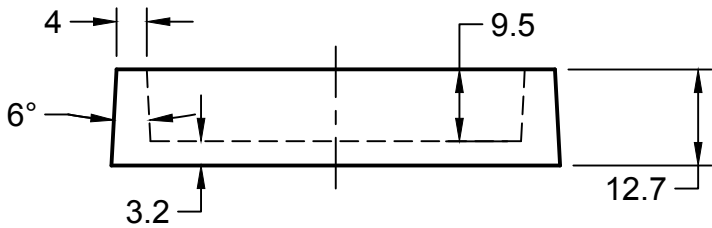
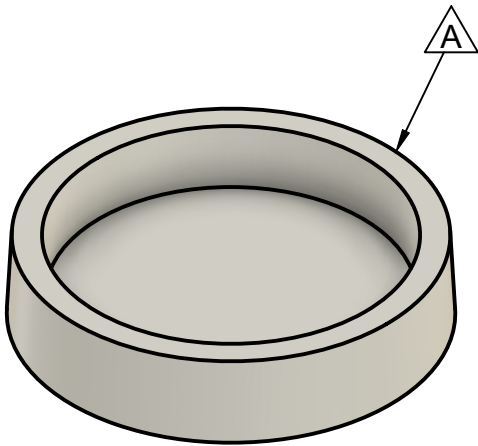
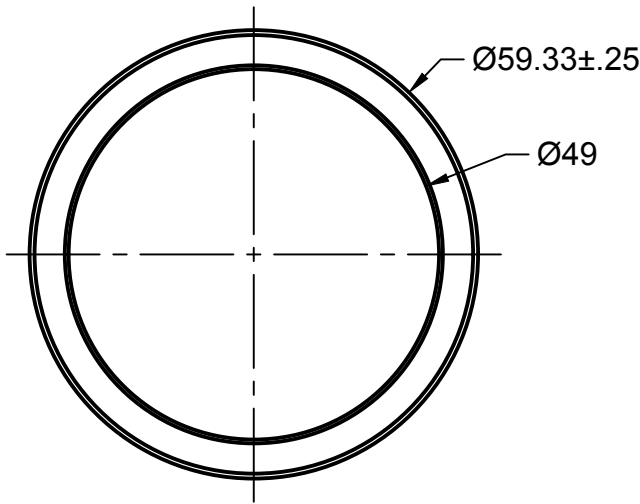
- NOTES:
1. REMOVE ALL BURRS AND SHARP EDGES.
 2. PART IS SYMMETRIC ABOUT CENTERLINES.

REVISION HISTORY			
REV	DESCRIPTION	DATE	APPROVED
A	1:0.9 SCALE DOWN	4/03/25	SK, CC
B	MATERIAL CHANGE FROM ALUMINUM TO HIGH DENSITY POLYETHYLENE	4/03/25	SK, CC
C	1:0.75 SCALE DOWN	4/17/25	SK, CC
D	THREAD ENGAGEMENT LENGTH INCREASED FROM 11 MM TO 13 MM	4/17/25	SK, CC
E	OVERALL THICKNESS INCREASED FROM 17 MM TO 19 MM	4/17/25	SK, CC
F	MATERIAL CHANGE FROM HIGH DENSITY POLYETHYLENE TO POLYPHENYLSULFONE	4/21/25	SK, CC
G	WINDOW THICKNESS DECREASED FROM 6 MM TO 3 MM	4/21/25	SK, CC
H	OVERALL THICKNESS INCREASED FROM 19 MM TO 24.5 MM	4/21/25	SK, CC
J	THREAD ENGAGEMENT LENGTH INCREASED FROM 19 MM TO 21.5 MM	4/21/25	SK, CC



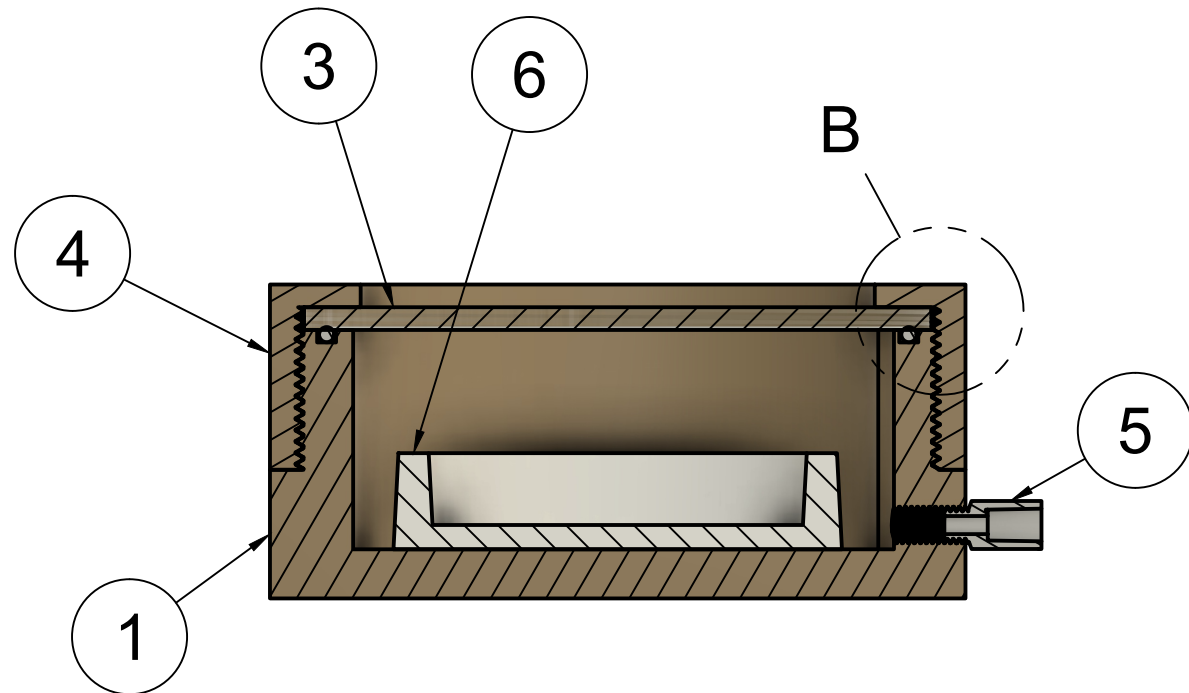
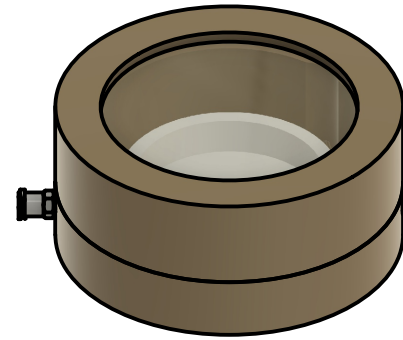
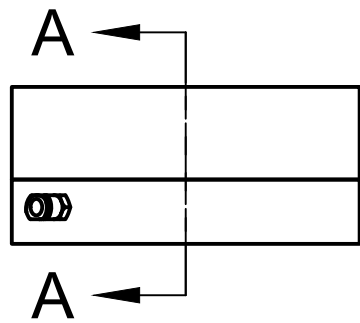
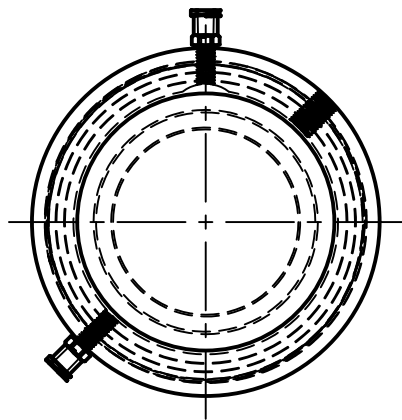
TOLERANCES UOS 1PL ±1 2PL ±.05 UNITS: METRIC	PROJECT BE-GY 6523 CAD Project				
	TITLE Cell Blast Apparatus CBA Rim				
APPROVED	SIZE	CODE	DWG NO	MATERIAL	REV
CHECKED Carolina Cen 4/21/25	B	ASME Y14.5	CBA-005	POLYPHENYLSULFONE (PPSU)	J
DRAWN Saba Khan 4/18/25	SCALE 1:1		WEIGHT 42.511 G		SHEET 5/7

REVISION HISTORY			
REV	DESCRIPTION	DATE	APPROVED
A	DIAMETER REDUCED FROM 75 MM TO 59.33 MM	4/17/25	SK, CC

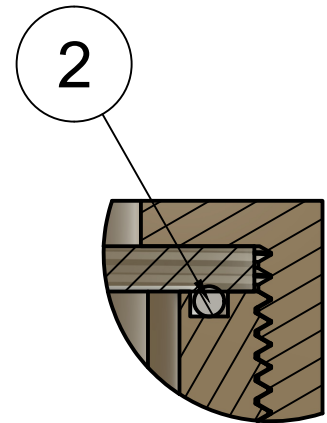


- NOTES:
- 1. PART INCLUDES 6° DRAFT FOR MOLD RELEASE.
 - 2. MAX OUTER DIAMETER SHALL NOT EXCEED Ø60 mm.

TOLERANCES UOS 1PL ±1 2PL ±.05 TOL ON ANGLE ±1° UNITS: METRIC	PROJECT BE-GY 6523 CAD Project					
	TITLE Cell Blast Apparatus PDMS Substrate					
APPROVED	SIZE	CODE		DWG NO	MATERIAL	REV
CHECKED Carolina Cen 4/21/25	B	ASME Y14.5		CBA-006	PDMS	A
DRAWN Saba Khan 4/18/25	SCALE 1:1			WEIGHT 17.009 G		SHEET 6/7



SECTION A-A
SCALE 1:1



DETAIL B
SCALE 2:1

PARTS LIST			
ITEM	QTY	PART NUMBER	MATERIAL
1	1	CBA BOTTOM	PEEK
2	1	O-RING	RUBBER, BUTYL
3	1	CBA TOP COVER	POLYCARBONATE, CLEAR
4	1	CBA RIM	PEEK
5	2	LUER INSERT	POLYCARBONATE, CLEAR
6	1	PDMS SUBSTRATE	ABS PLASTIC

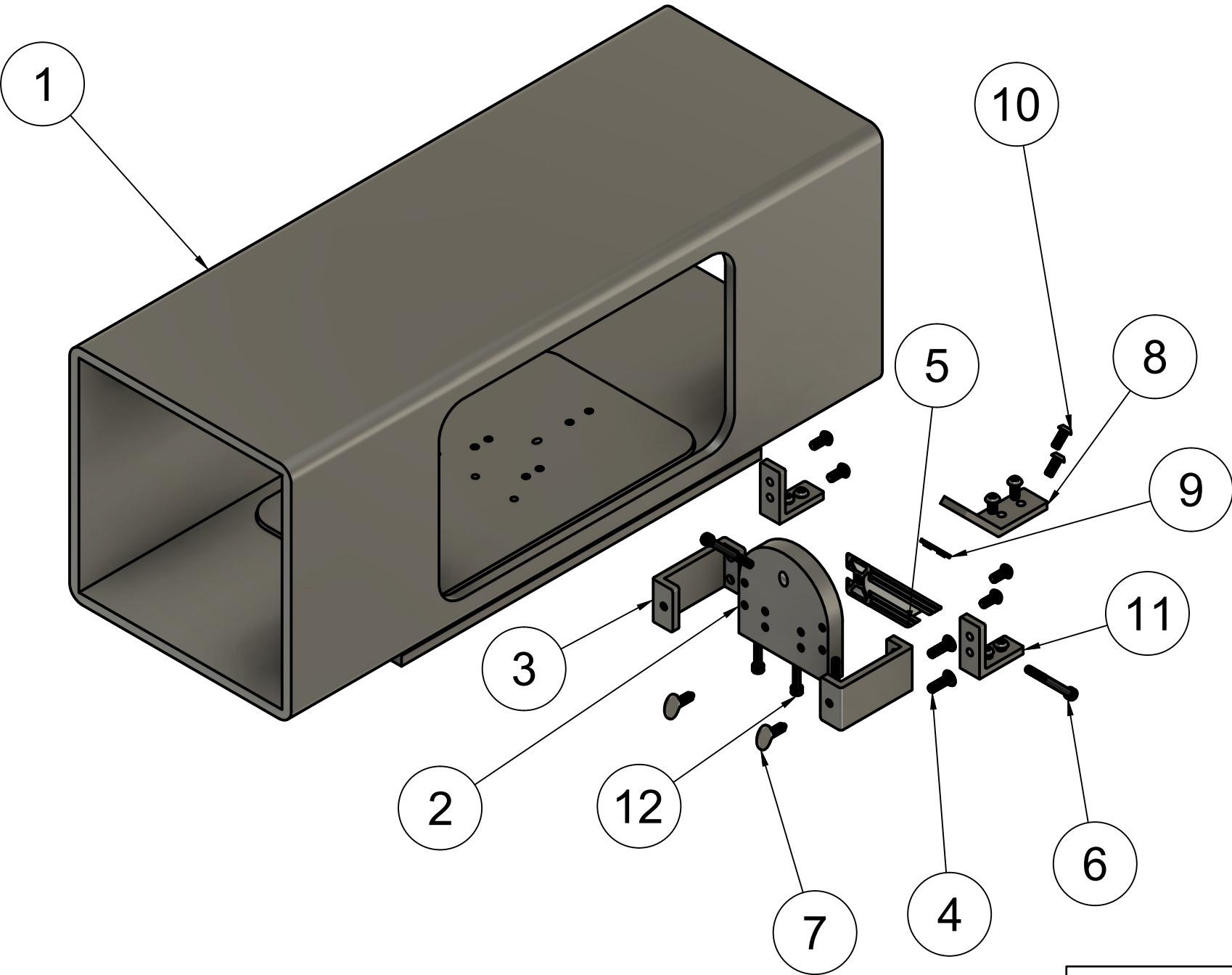
NOTES:

- ALL PARTS TO BE ASSEMBLED PER EXPLODED VIEW.
- ALL COMPONENTS MUST BE CLEANED AND DRY PRIOR TO ASSEMBLY.

	PROJECT					
	BE-GY 6523 CAD Project					
	TITLE					
Cell Blast Apparatus						
CBA Assembly						
APPROVED			SIZE	CODE	DWG NO	REV
CHECKED	Carolina Cen	4/21/25	B	ASME Y14.5	CBA-007	N/A
DRAWN	Saba Khan	4/18/25	SCALE 1:2		WEIGHT 241.696 G	SHEET 7/7

Mounting Apparatus

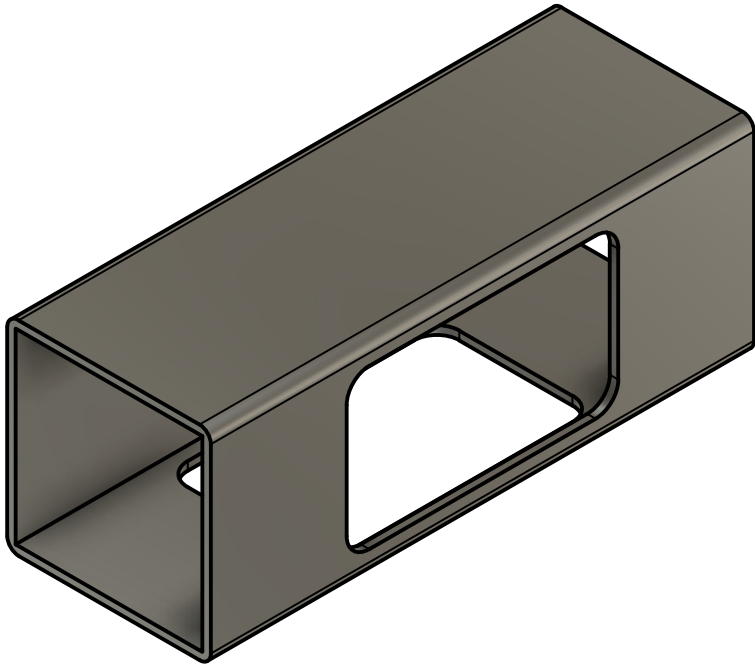
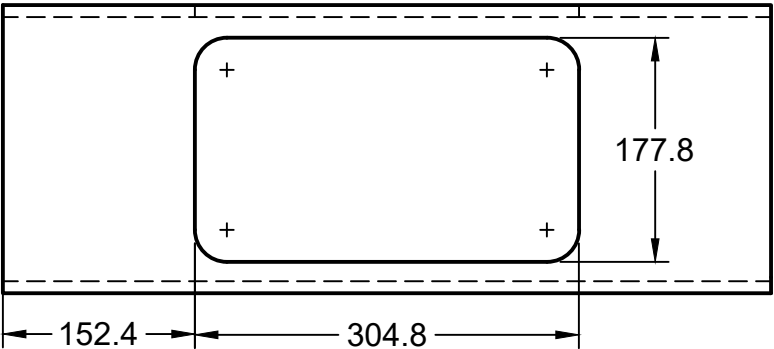
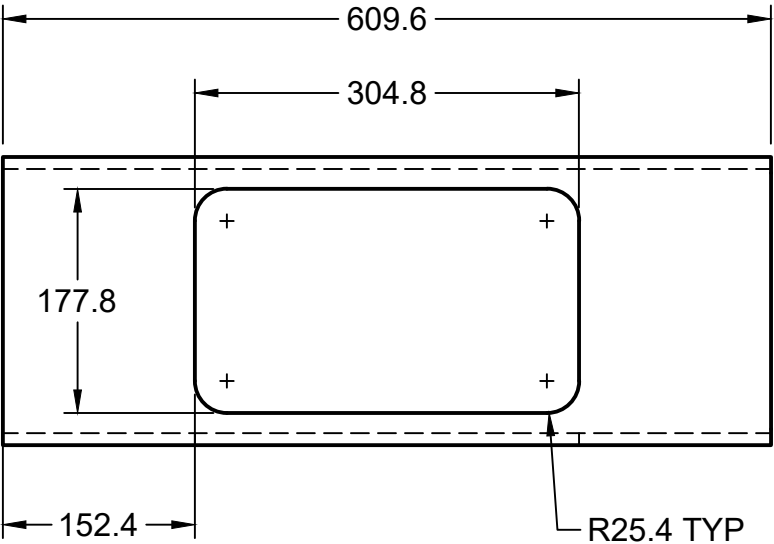
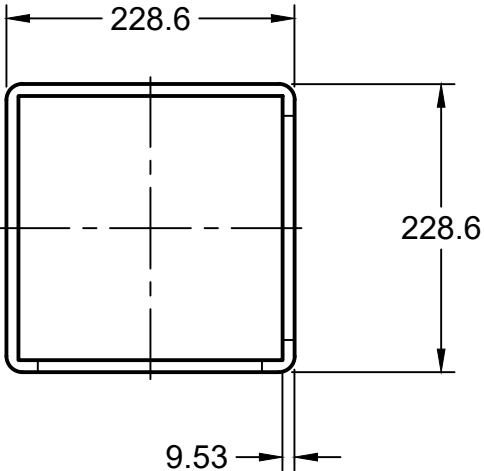
2D Drawing



PARTS LIST				
ITEM	QTY	PART NUMBER	MATERIAL	MASS
1	1	BLAST TUBE ASSEMBLY V7	STEEL AISI 1045 450 QT	42557.954 G
2	1	HEADBOARD FOR MA	STEEL AISI 1045 450 QT	872.14 G
3	2	C BRACKET UPDATED	STEEL AISI 1045 450 QT	190.841 G
4	4	C BRACKET TO HEADBOARD SCREW	---- BLACK OXIDE ALLOY STEEL	0.00 G
5	1	45 DEGREE T SLOT	ALUMINUM 6061	2.05 OUNCEMASS
6	2	HEADBOARD-BASEP LATE TO TSLOT SCREW	BLACK OXIDE ALLOY STEEL	0.00 G
7	2	90772A537_SOFT-TI P THUMB SCREW	..NYLON TIP, ZINC PLATED STEEL	0.22 OUNCEMASS
8	1	45DEG_BRACKET	STEEL AISI 1045 450 QT	111.251 G
9	1	5537T464_T-SLOTTE D FRAMING	ZINC PLATED STEEL	7.34 G
10	12	92095A227_BUTTON HEAD HEX DRIVE SCREW	STEEL PASSIVATED 18-8 STAINLESS STEEL	3.922 G
11	2	90DEG_BRACKET	STEEL AISI 1045 450 QT	75.67 G
12	3	HEADBOARD TO BASEBOARD SCREW 2	STEEL, ALLOY	8.173 G

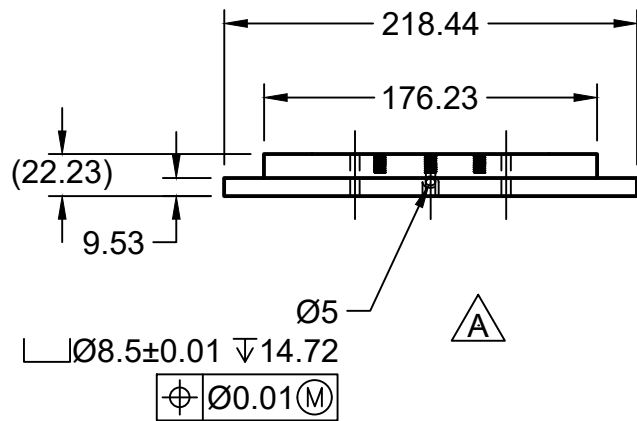
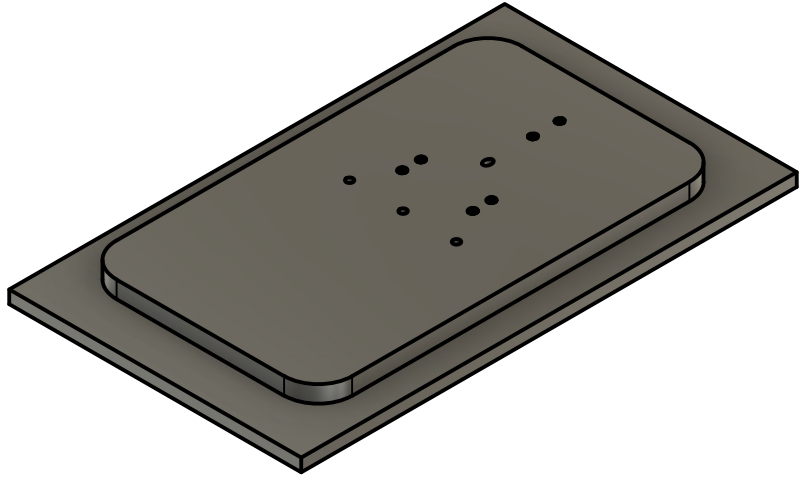
	PROJECT			
	BE-GY 6523 CAD Project			
	TITLE			
Mounting Apparatus Final Design Bill of Material				
APPROVED		SIZE	CODE	DWG NO
CHECKED Saba Khan 4/21/2025		B	ASME Y14.5	MA-001
DRAWN Carolina Cen 4/19/2025		SCALE 1:4		REV N/A
		WEIGHT 44220.06 G		SHEET 1/8

NOTE: PER DESIGN REQUIREMENT
OUTLINED BY PROFESSOR HANNA



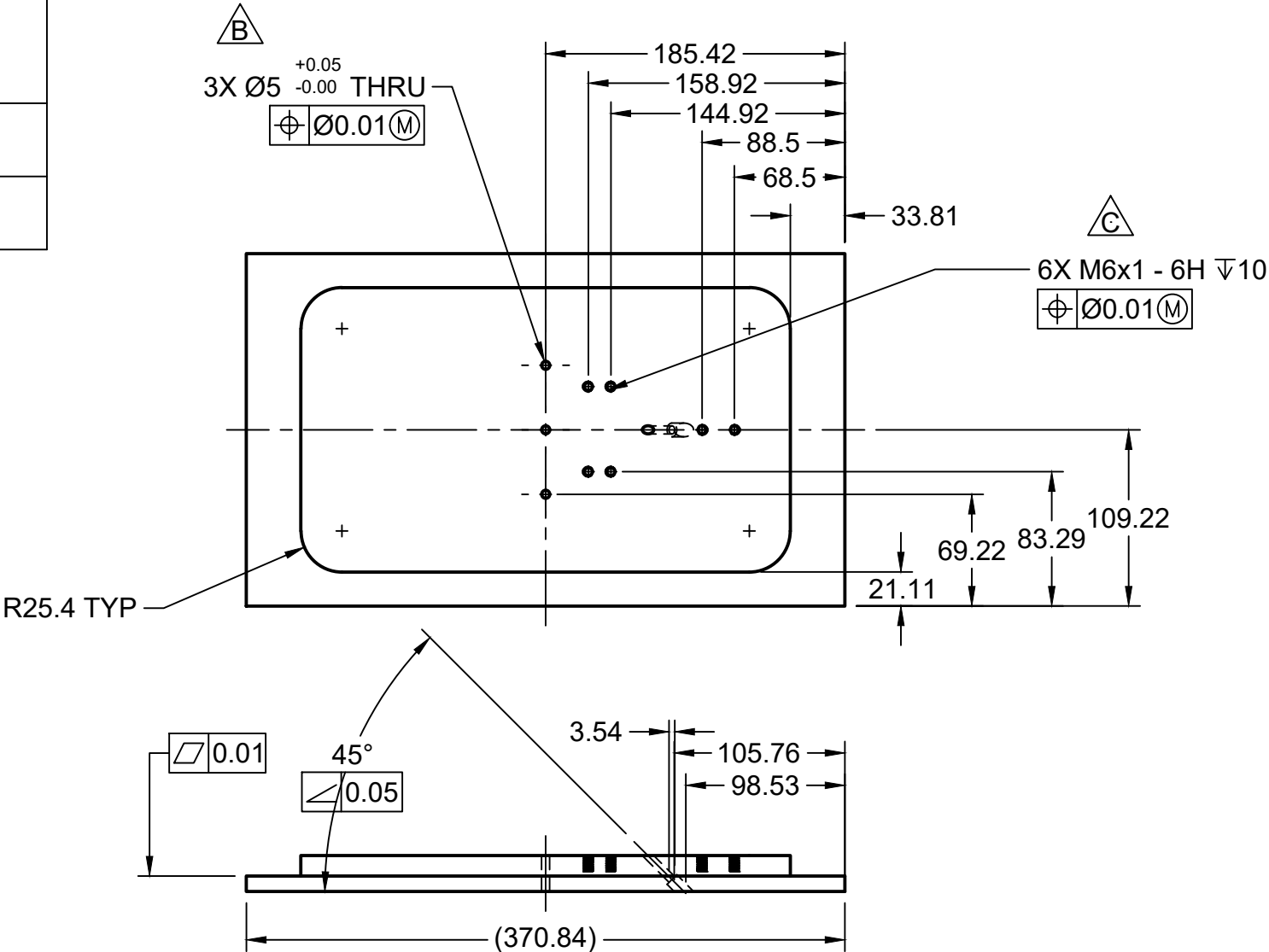
TOLERANCES UOS 1PL ±0.8 2PL ± 0.5 3PL ± 0.1 UNITS: METRIC	PROJECT BE-GY 6523 CAD Project					
	TITLE Mounting Apparatus Final Design Blast Tube					
APPROVED	SIZE	CODE		DWG NO	MATERIAL: STEEL AISI	REV
CHECKED Saba Khan 4/21/2025	B	ASME Y14.5		MA-002	1045 450 QT	1
DRAWN Carolina Cen 4/19/2025	SCALE 1:6			WEIGHT 31258.29 G		SHEET 2/8

REVISION HISTORY			
REV	DESCRIPTION	DATE	APPROVED
A	1 Ø5 CLEARANCE HOLE OF COUNTERBORE Ø8.5 WAS ADDED FOR THE SCREW CONNECTING THE BASE PLATE TO THE 45° T-SLOT	4/3/2025	CC, SK
B	3 CLEARANCE HOLES OF Ø5 WERE ADDED	4/17/2025	CC, SK
C	6 THREAD HOLES OF M6 X 1 FOR 45°AND 90° BRACKETS WERE ADDED	4/17/2025	CC, SK

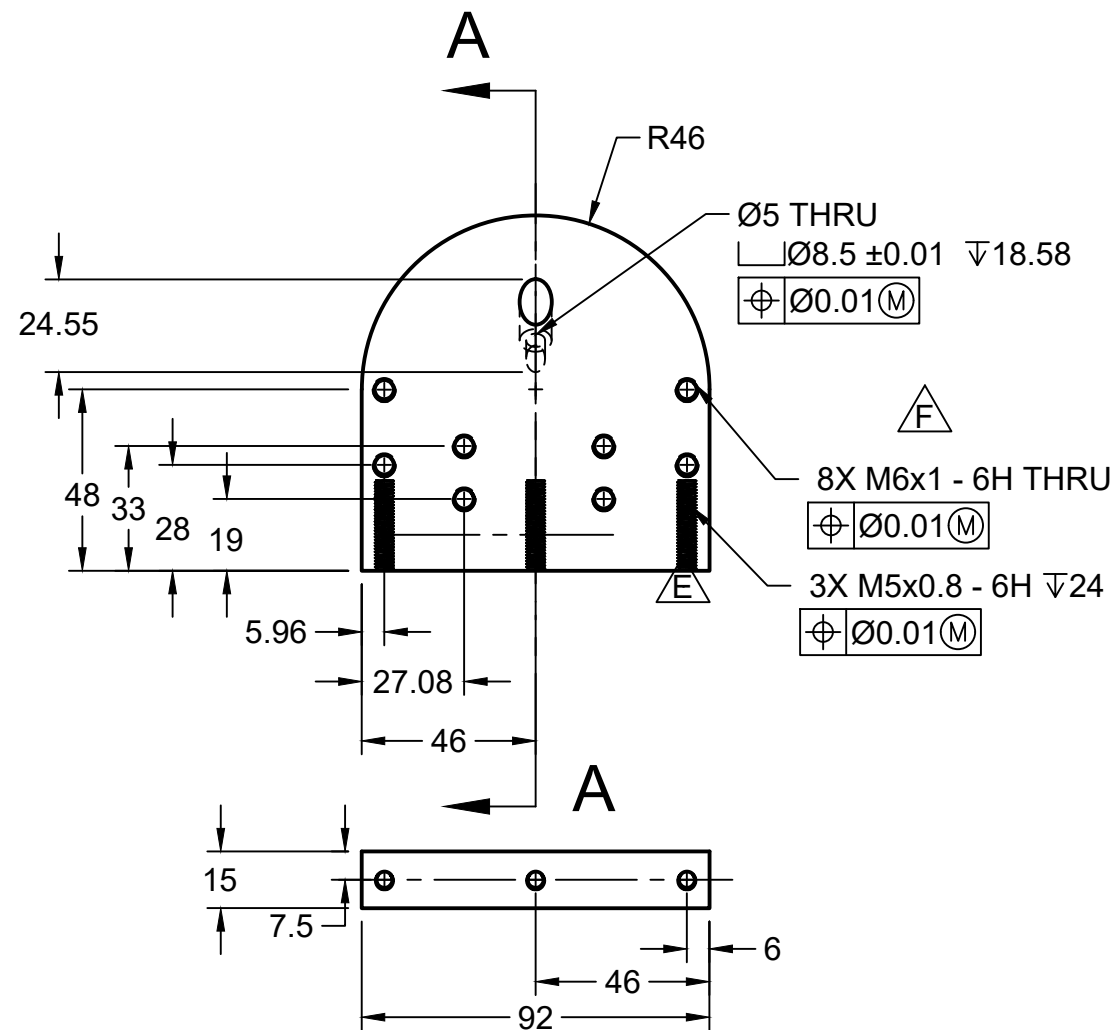
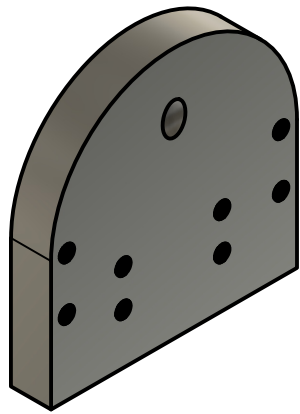


NOTES:

1. REMOVE ALL BURRS AND SHARP
2. STANDARD PASSIVATED 18-8 STAINLESS STEEL, M6 X 1MM BUTTON HEAD HEX DRIVE SCREW, TO BE FIT IN THE M6X1-6H HOLES
3. STANDARD M5 X 0.8MM PARTIALLY THREADED SOCKET HEAD SCREW OF 60MM LONG, TO BE FIT IN THE M5 X 0.8MM -6H HOLES - CONNECTING THE BOTTOM PLATE TO A 45° T-SLOT
3. STANDARD M5 X 0.8MM PARTIALLY THREADED SOCKET HEAD SCREW OF 60MM LONG, TO BE FIT IN THE M5 X 0.8MM -6H HOLES - CONNECTING THE BOTTOM PLATE TO A HEADBOARD

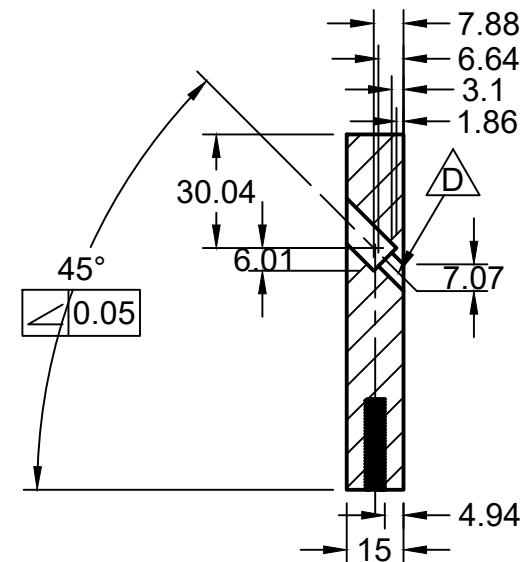


TOLERANCES UOS 1PL ± 0.1 2PL ± 0.05 ANGLE ± 1° UNITS: METRIC	PROJECT BE-GY 6523 CAD Project				
	TITLE Mounting Apparatus Final Design Bottom Plate				
APPROVED	SIZE	CODE	DWG NO	MATERIAL: STEEL AISI	REV
CHECKED SABA KHAN 4/21/2025	B	ASME Y14.5	MA-002	1045 450 QT	C
DRAWN Carolina Cen 4/19/2025	SCALE 1:4		WEIGHT 11299 G		SHEET 3/8



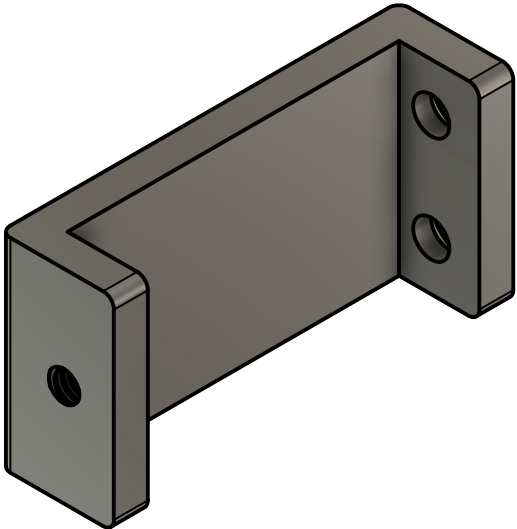
- NOTES:
1. REMOVE ALL BURRS AND SHARP
 2. PART IS SYMMETRIC ABOUT CENTERLINES
 3. STANDARD M5 X 0.8MM PARTIALLY THREADED SOCKET HEAD SCREW OF 45MM LONG, TO BE FIT IN THE M5 X 0.8MM -6H HOLES - CONNECTING TO THE BOTTOM PLATE
 4. STANDARD PASSIVATED 18-8 STAINLESS STEEL, M6 X 1MM BUTTON HEAD HEX DRIVE SCREW, TO BE FIT IN THE M6X1-6H HOLES
 5. STANDARD M5 X 0.8MM PARTIALLY THREADED SOCKET HEAD SCREW OF 60MM LONG, TO BE FIT IN THE Ø8.5 COUNTERBORE HOLE - CONNECTING TO A 45° T-SLOT

REVISION HISTORY			
REV	DESCRIPTION	DATE	APPROVED
A	CURVE OF R46 WAS ADDED TO REMOVE THE SHARP EDGES OF THE RECTANGULAR BOARD	4/3/2025	CC, SK
B	1:0.75 SCALE DOWN	4/17/2025	CC, SK
C	INCREASED THICKNESS FROM 9 TO 15 MM	4/17/2025	CC, SK
D	1 Ø5 CLEARANCE HOLE OF COUNTERBORE Ø8.5 WAS ADDED FOR THE SCREW CONNECTING TO THE 45° T-SLOT	4/17/2025	CC, SK
E	3 CLEARANCE HOLES OF Ø5 WERE ADDED	4/17/2025	CC, SK
F	4 THREAD HOLES OF M6 X 1 FOR TWO 90° BRACKETS WERE ADDED	4/17/2025	CC, SK

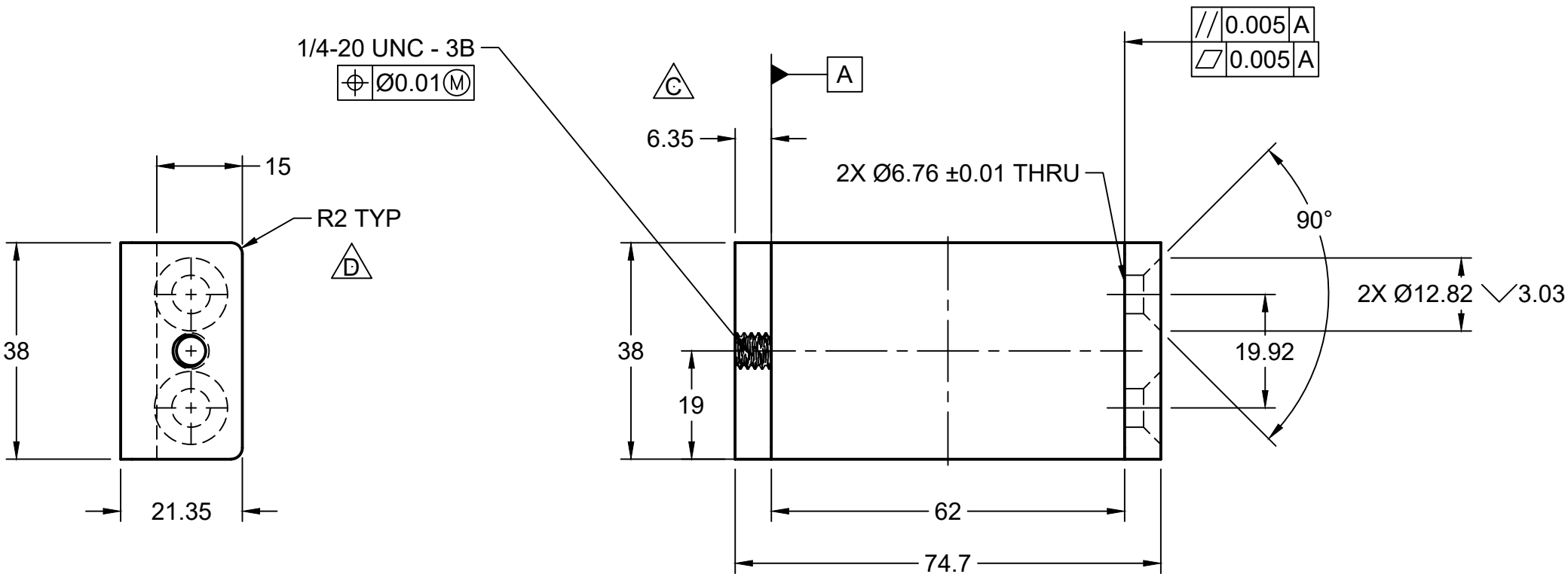


SECTION A-A
SCALE 1:2

TOLERANCES UOS 1PL ± 0.1 2PL ± 0.05 ANGLE ± 1° UNITS: METRIC		PROJECT BE-GY 6523 CAD Project			
		TITLE Mounting Apparatus Final Design Headboard			
APPROVED	SIZE	CODE	DWG NO	MATERIAL: STEEL AISI	REV
CHECKED Saba Khan 4/21/2025	B	ASME Y14.5	MA-004	1045 450 QT	F
DRAWN Carolina Cen 4/19/2025	SCALE 1:2		WEIGHT 872.140 G		SHEET 4/8

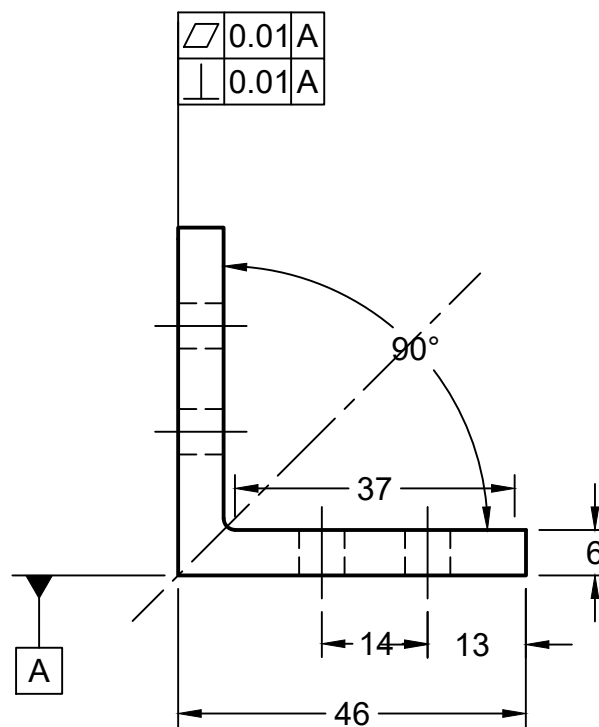
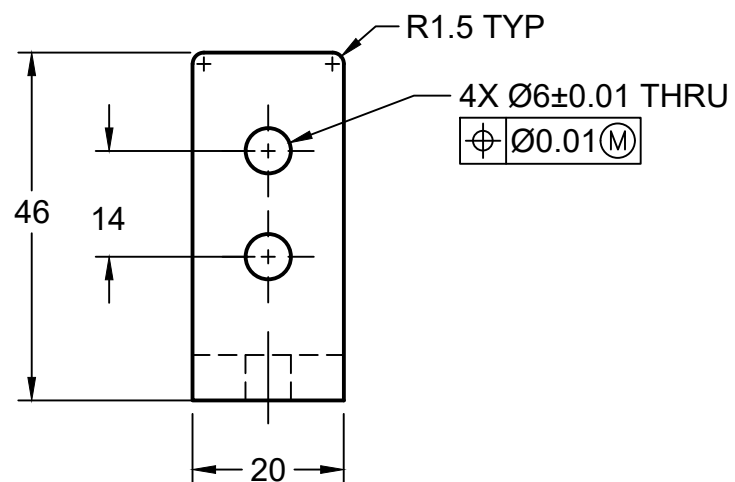
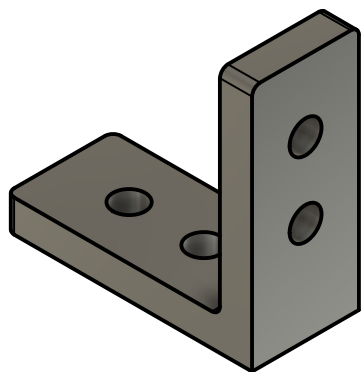


REVISION HISTORY			
REV	DESCRIPTION	DATE	APPROVED
A	1:09 SCALE DOWN	4/3/2025	CC, SK
B	1:075 SCALE DOWN	4/17/2025	CC, SK
C	INCREASED THICKNESS TO 6.35	4/17/2025	CC, SK
D	FILLET OF R2 WAS ADDED TO THE CORNERS TO REMOVE HAVING SHARP EDGES	4/17/2025	CC, SK



- NOTES:
- REMOVE ALL BURRS AND SHARPS
 - STANDARD HEX DRIVE FLAT HEAD SCREW, 90° CCW, M6X1 THREAD, TO BE INSERTED IN THE M6x1-6H HOLES - TO BE MOUNTED ONTO THE HEADBOARD
 - STANDARD 1/4-20 UNF SOFT-TIP THUMB SCREW TO BE INSERTED IN THE 1/4-20 UNC-3B HOLE - SECURE THE CBA COMPONENT AGAINST THE HEADBOARD

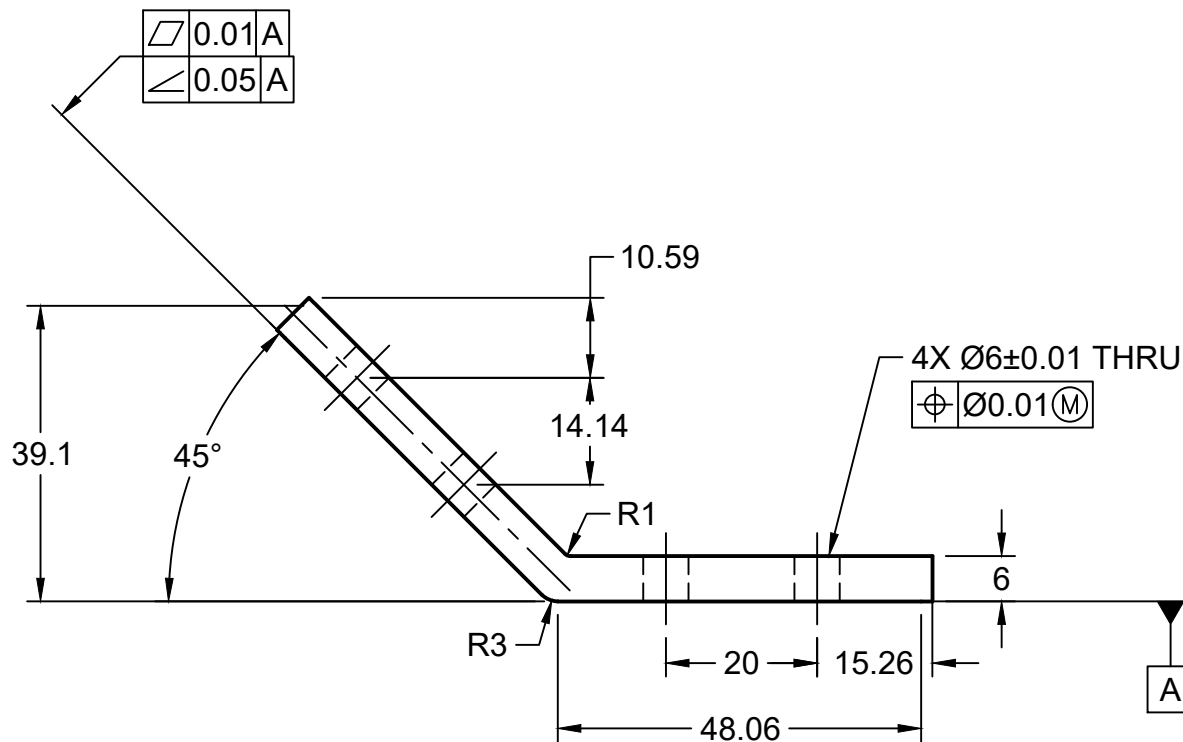
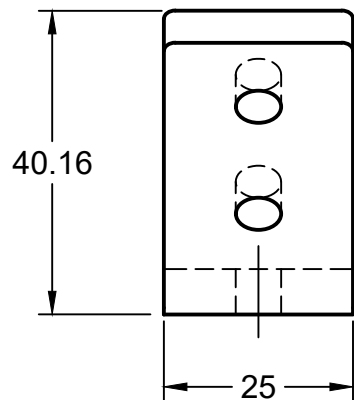
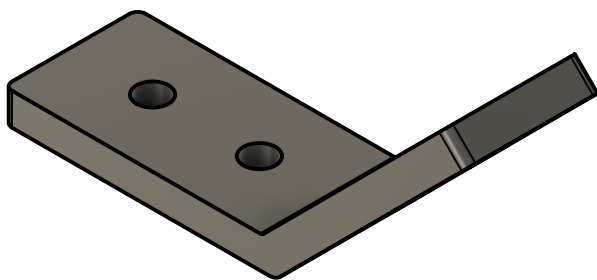
TOLERANCES UOS 1PL ± 0.1 2PL ± 0.05 ANGLE $\pm 1^\circ$ UNITS: METRIC		PROJECT BE-GY 6523 CAD Project			
		TITLE Mounting Apparatus Final Design C-Brackets			
APPROVED	SIZE	CODE	DWG NO	MATERIAL: STEEL AISI	REV
CHECKED Saba Khan 4/21/2025	B	ASME Y14.5	MA-005	1045 450 QT	D
DRAWN Carolina Cen 4/21/2025	SCALE 1:1		WEIGHT 190.841 G	SHEET 5/8	



NOTES:

1. REMOVE ALL BURRS AND SHARPS
2. PART IS SYMMETRIC ABOUT CENTERLINES
3. STANDARD PASSIVATED 18-8, M6 X 1MM THREAD BUTTON HEAD HEX DRIVE SCREW, 14MM LONG, TO BE INSERTED IN THE Ø6 HOLES - CONNECTING THE HEADBOARD AND BOTTOM PLATE

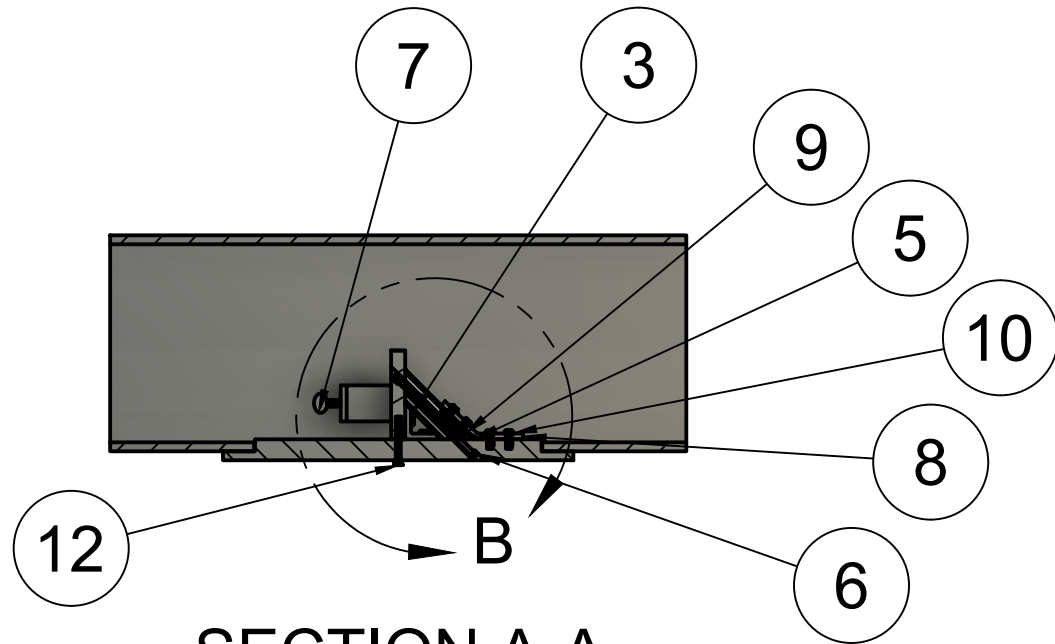
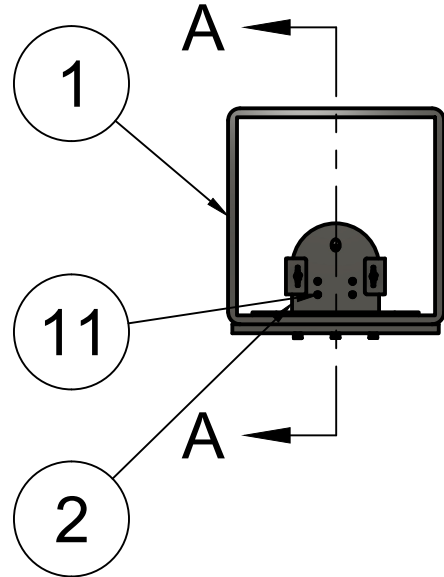
TOLERANCES UOS 1PL ± 0.1 2PL ± 0.05 ANGLE ± 1° UNITS: METRIC		PROJECT BE-GY 6523 CAD Project			
		TITLE Mounting Apparatus Final Design 90 Degree Bracket			
APPROVED		SIZE	CODE	DWG NO	MATERIAL: STEEL AISI
CHECKED Saba Khan 4/21/2025		B	ASME Y14.5	MA-006	1045 450 QT
DRAWN Carolina Cen 4/19/2025		SCALE 1:1		WEIGHT 75.670 G	SHEET 6/8
				REV	1



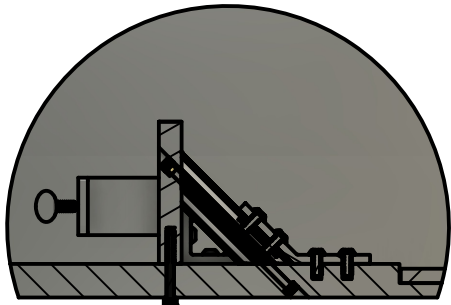
NOTES:

1. REMOVE ALL BURRS AND SHARPS
2. PART IS SYMMETRIC ABOUT CENTERLINES
3. STANDARD PASSIVATED 18-8, M6 X 1MM THREAD BUTTON HEAD HEX DRIVE SCREW, 14MM LONG, TO BE INSERTED IN THE Ø6 HOLES - CONNECTING THE HEADBOARD AND BOTTOM PLATE

TOLERANCES UOS 1PL ± 0.1 2PL ± 0.05 ANGLE ± 1° UNITS: METRIC			PROJECT BE-GY 6523 CAD Project			
			TITLE Mounting Apparatus Final Design 45 Degree Bracket			
APPROVED			SIZE	CODE	DWG NO	MATERIAL: STEEL AISI
CHECKED Saba Khan 4/21/2025			B	ASME Y14.5	MA-007	1045 450 QT
DRAWN Carolina Cen 4/19/2025			SCALE 1:1		WEIGHT 111.251 G	SHEET 7/8
					REV	1



SECTION A-A
SCALE 1:8



DETAIL B
SCALE 1:5

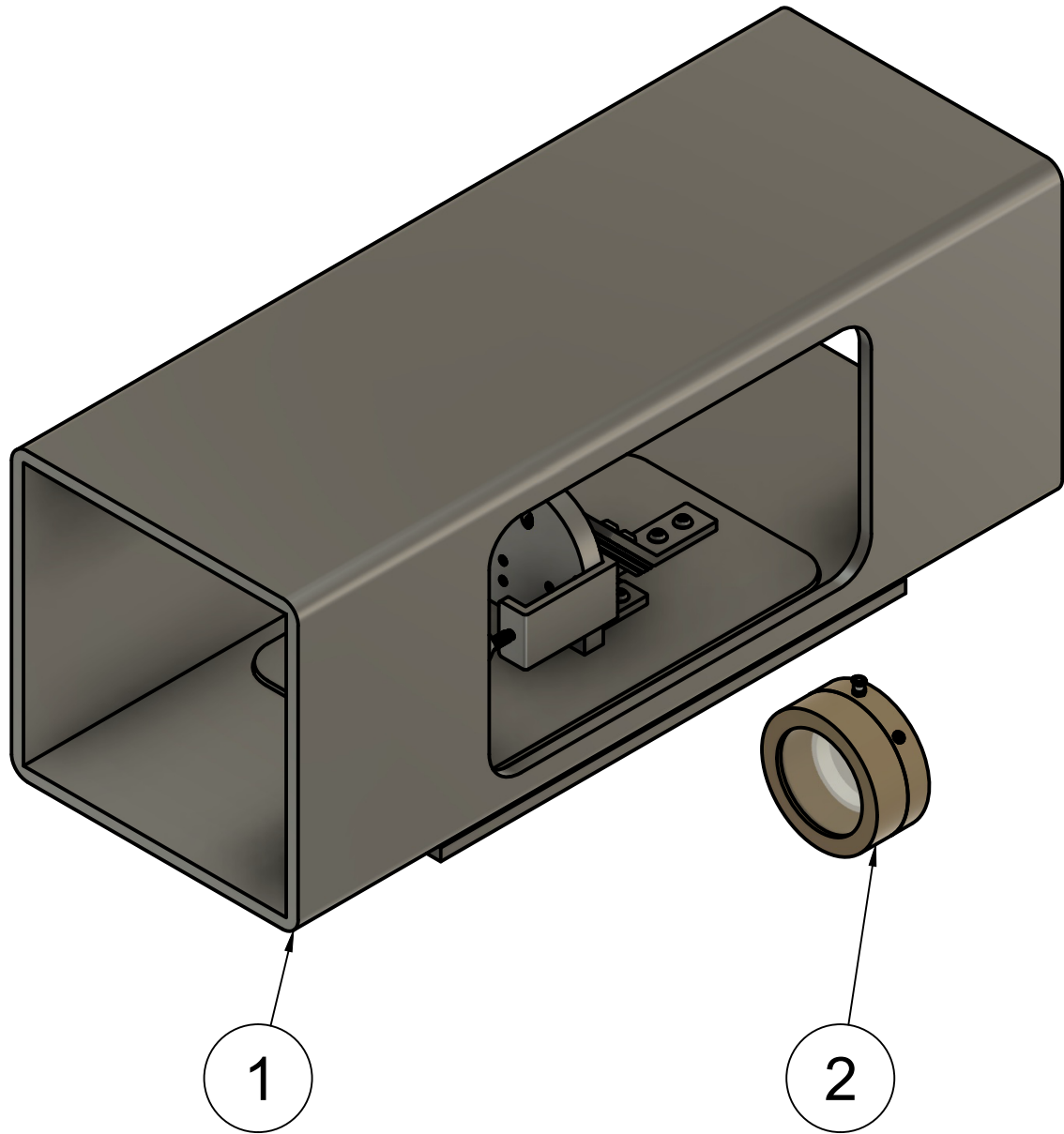
PARTS LIST			
ITEM	QTY	PART NUMBER	MATERIAL
1	1	BLAST TUBE ASSEMBLY V7	STEEL AISI 1045 450 QT
2	1	HEADBOARD FOR MA	STEEL AISI 1045 450 QT
3	2	C BRACKET UPDATED	STEEL AISI 1045 450 QT
4	4	C BRACKET TO HEADBOARD SCREW	----BLACK OXIDE ALLOY STEEL
5	1	45 DEGREE T SLOT	ALUMINUM 6061
6	2	HEADBOARD-BASEP LATE TO TSLOT SCREW	----BLACK OXIDE ALLOY STEEL
7	2	90772A537_SOFT-TI P THUMB SCREW	NYLON TIP, ZINC PLATED STEEL
8	1	45DEG_BRACKET	STEEL AISI 1045 450 QT
9	1	5537T464_T-SLOTTE D FRAMING	ZINC PLATED STEEL
10	12	92095A227_BUTTON HEAD HEX DRIVE SCREW	STEEL PASSIVATED 18-8 STAINLESS STEEL
11	2	90DEG_BRACKET	STEEL AISI 1045 450 QT
12	3	HEADBOARD TO BASEBOARD SCREW 2	STEEL, ALLOY

NOTES:

- ALL PARTS TO BE ASSEMBLED PER EXPLODED VIEW.
- ALL COMPONENTS MUST BE CLEANED AND DRY PRIOR TO ASSEMBLY.

		PROJECT		
		BE-GY 6523 CAD Project		
		TITLE		
		Mounting Apparatus Final Design MA Assembly		
APPROVED		SIZE	CODE	DWG NO
CHECKED Saba Khan 4/21/2025		B	ASME Y14.5	MA-008
DRAWN Carolina Cen 4/19/2025		SCALE 1:8		REV N/A
		WEIGHT 44220.06 G		SHEET 8/8

Full Assembly 2D *Drawing*

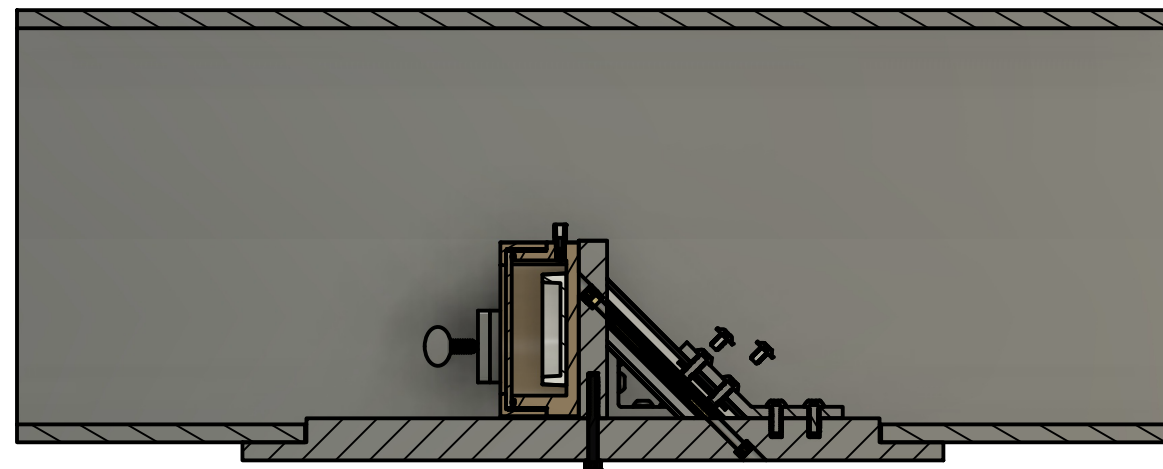
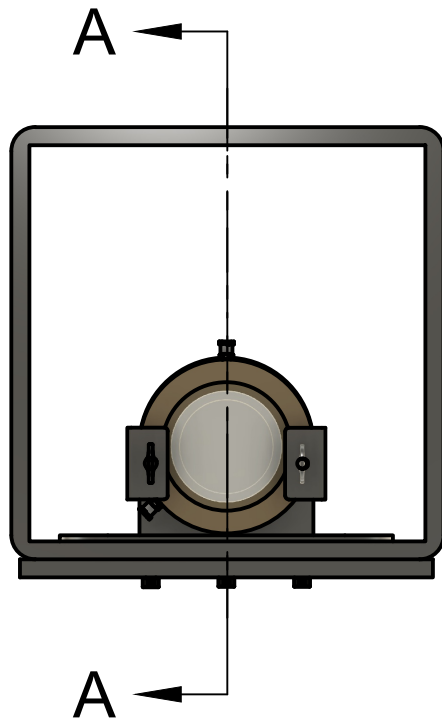


PARTS LIST			
ITEM	QTY	PART NUMBER	MATERIAL
1	1	MOUNTING APPARATUS FINAL DESIGN	----
2	1	CBA FINAL DESIGN	----

- NOTES:
- 1. PART 1 DESIGN SPECIFICATIONS OUTLINED IN MA-00#
 - 2. PART 2 DESIGN SPECIFICATIONS OUTLINED IN CBA-00#

REVISION HISTORY			
REV	DESCRIPTION	DATE	APPROVED
A	REDUCED THE CROSS SECTIONAL AREA FROM 80% TO 30%	4/19/25	SK, CC
B	REDUCED THE CROSS SECTIONAL AREA FROM 30% TO 20%	4/19/25	SK, CC

		PROJECT			
		BE-GY 6523 CAD Project			
		TITLE			
		Full Assembled Design			
		APPROVED	SIZE	CODE	DWG NO
CHECKED		Carolina Cen	4/21/24	B	ASME Y14.5
DRAWN		Saba Khan	4/19/25	SCALE 1:4	WEIGHT N/A
				SHEET 1/2	REV B



SECTION A-A
SCALE 1:4

- NOTES:
- CROSS SECTIONAL VIEW OF THE ENTIRE ASSEMBLY INCLUDING THE CELL BLAST APPARTUS AND THE MOUNTING APPARATUS.

			PROJECT			
			BE-GY 6523 CAD Project			
			TITLE			
			Full Assembled Design			
APPROVED			SIZE	CODE	DWG NO	REV
CHECKED	Carolina Cen	4/21/24	B	ASME Y14.5	FA-001	B
DRAWN	Saba Khan	4/19/25	SCALE 1:8		WEIGHT N/A	SHEET 2/2